

Langchain all models in GEN AI

```
# =====  
# 🧠 1. PROMPTS  
# =====  
  
# Old (still used)  
from langchain.prompts import (  
    PromptTemplate,  
    ChatPromptTemplate,  
    FewShotPromptTemplate,  
    FewShotChatMessagePromptTemplate,  
    MessagesPlaceholder,  
    SystemMessagePromptTemplate,  
    HumanMessagePromptTemplate,  
    AIMessagePromptTemplate  
)  
  
# New (Recommended)  
from langchain_core.prompts import (  
    PromptTemplate,  
    ChatPromptTemplate,  
    MessagesPlaceholder  
)
```

```
# =====  
# 🤖 2. LLMs & CHAT MODELS  
# =====  
  
from langchain_openai import ChatOpenAI, OpenAI  
from langchain_community.llms import HuggingFaceHub  
from langchain_community.chat_models import ChatOllama  
  
# =====  
# 🔗 3. RUNNABLES (NEW)  
# =====  
  
from langchain_core.runnables import (  
    RunnableSequence,  
    RunnableLambda,  
    RunnableParallel,  
    RunnablePassthrough  
)  
  
# Old Chains (important for interviews)  
from langchain.chains import (  
    LLMChain,  
    SequentialChain,  
    SimpleSequentialChain  
)
```

```
# =====  
# 🧠 4. MEMORY  
# =====
```

```
from langchain.memory import (  
    ConversationBufferMemory,  
    ConversationSummaryMemory,  
    ConversationBufferWindowMemory  
)
```

```
# =====  
# 📄 5. DOCUMENT LOADERS  
# =====
```

```
from langchain_community.document_loaders import (  
    PyPDFLoader,  
    TextLoader,  
    WebBaseLoader,  
    DirectoryLoader,  
    CSVLoader  
)
```

```
# =====  
# ✂ 6. TEXT SPLITTERS  
# =====
```

```
from langchain.text_splitter import (  
    RecursiveCharacterTextSplitter,  
    CharacterTextSplitter  
)
```

```
# =====  
# 📄 7. EMBEDDINGS  
# =====
```

```
from langchain_openai import OpenAIEmbeddings  
from langchain_community.embeddings import HuggingFaceEmbeddings
```

```
# =====  
# 🧠 13. MESSAGES / SCHEMA  
# =====
```

```
from langchain_core.messages import (  
    HumanMessage,  
    AIMessage,  
    SystemMessage  
)
```

```

# =====
# 📁 8. VECTOR STORES
# =====

from langchain_community.vectorstores import (
    FAISS,
    Chroma
)

# External
from langchain_pinecone import PineconeVectorStore

# =====
# 🔍 9. RETRIEVERS
# =====

from langchain_core.vectorstores import VectorStoreRetriever
from langchain.retrievers import (
    MultiQueryRetriever,
    ContextualCompressionRetriever
)

```

```

# =====
# 🤖 10. AGENTS
# =====

from langchain.agents import (
    initialize_agent,
    AgentExecutor,
    create_openai_tools_agent
)

# =====
# 🛠️ 11. TOOLS
# =====

from langchain.tools import Tool
from langchain_core.tools import tool

# =====
# 🌸 12. OUTPUT PARSERS
# =====

from langchain_core.output_parsers import (
    StrOutputParser,
    JsonOutputParser
)

```

```
# =====  
# 🔥 MOST IMPORTANT (MUST KNOW)  
# =====  
  
from langchain_core.prompts import ChatPromptTemplate  
from langchain_openai import ChatOpenAI  
from langchain_core.runnables import RunnableSequence  
from langchain_core.output_parsers import StrOutputParser  
  
# =====  
# 🚀 MODERN PIPELINE (NEW STANDARD)  
# =====  
  
from langchain_core.prompts import ChatPromptTemplate  
from langchain_openai import ChatOpenAI  
from langchain_core.output_parsers import StrOutputParser  
  
prompt = ChatPromptTemplate.from_template("Explain {topic}")  
model = ChatOpenAI()  
parser = StrOutputParser()  
  
chain = prompt | model | parser  
  
response = chain.invoke({"topic": "Transformers in NLP"})  
print(response)
```